Dosesure: Smart Medicine Dispenser

Om Santosh Kamble
Department of Electronic & Instrumentation Science
Savitribai Phule Pune University
Pune, India

Email: omkamble1231@gmail.com

Denalven Sawian
Department of Electronic & Instrumentation Science
Savitribai Phule Pune University
Pune, India

Email: denalvenmark18@gmail.com

Abstract— Medication non-adherence is a major global healthcare challenge, particularly for patients undergoing long-term treatment for chronic diseases such as tuberculosis, diabetes, and cardiovascular disorders. Existing reminder-based systems lack mechanisms to verify actual medication intake, resulting in unreliable adherence records. This paper presents a smart medicine dispenser integrating time-based reminders, load-cell based intake verification, GSM alerts, and cloud-based adherence logging. An ESP32 microcontroller interfaced with HX711 load-cell amplifiers verifies tablet removal, while Google Sheets enables real-time remote monitoring. Experimental evaluation demonstrates reliable intake detection, accurate scheduling, and robust communication, making the system suitable for home and remote healthcare applications.

Keywords—Medication adherence, Smart Medicine Dispenser, ESP32, Loadcell, GSM

I. Introduction

Adherence to medication schedules is a critical factor in the success of long-term treatments for chronic diseases such as tuberculosis, epilepsy, and cardiac disorders. Missing doses can lead to severe complications, yet patients—particularly the elderly or those with cognitive impairments—often struggle to follow complex regimens. The development of automated medication management has evolved significantly over the past two decades through various technological phases. Early foundational research by [1] and [2] established initial frameworks for remote oversight and IoT-based pill reminders, while [3] explored the use of audiovisual behaviour prompting (AVRD) systems to guide patient actions. As the field progressed, research shifted toward more active communication and localized sensing; [4] integrated GSM modules with ultrasonic sensors for presence detection, and [5] utilized Bluetooth Low Energy for local scheduling. Later, [6] introduced specialized features such as biometric access and cooling systems, while more recent work by [7], [8], and [9] has focused on multi-compartment units using IR sensors and web-based monitoring interfaces. Furthermore, [10] and [11] have emphasized the growing need for physical verification to bridge the gap between a device dispensing a pill and the patient actually receiving it.

Despite these technological milestones, existing literature reveals several critical drawbacks that limit the clinical reliability and universal adoption of current dispensers. A primary limitation is the "verification gap" inherent in optical sensing; systems relying on IR or ultrasonic sensors can be triggered by ambient light changes, dust, or accidental movement, which does not accurately confirm the removal of medication. Furthermore, many modern IoT dispensers, such as those described by [7], rely heavily on manual patient confirmation—a significant failure point for memory-impaired users. There is also a notable lack of scalability in previous designs; many require specialized, proprietary software or local-only connectivity
(Bluetooth/Zigbee), making it difficult for caregivers to monitor patients across different platforms. Finally, most traditional systems operate as isolated dispensing units, failing to provide a centralized, real-time data stream that correlates medication events with the patient's broader physiological state, leaving a critical data gap for healthcare providers.

The proposed Dosesure: Smart Medicine Dispenser & Health Monitoring System is a comprehensive autonomous platform designed for daily multi-dose management. The hardware is organized into a three-slot configuration for morning, afternoon, and evening cycles, managed by an ESP32 microcontroller and a DS3231 Real-Time Clock (RTC) to ensure precise timing independent of internet fluctuations. At scheduled intervals, the system issues proactive GSM-based SMS reminders via a SIM800L module. To ensure patient safety, the system implements an active "Wrong Compartment" protocol; if a user attempts to access a medication slot out of its designated sequence, the device immediately triggers an auditory buzzer alert and logs the violation to the cloud. This dual-layered approach—combining local safety mechanisms with a scalable, cloud-integrated infrastructure—provides a robust solution for remote healthcare oversight, allowing medical professionals to monitor adherence trends and physiological metrics in real-time

II. Literature Review

The early phase of automated medication management was defined by a transition from static pill organizers to connected electronic systems. [1] and [2] pioneered the use of central servers and basic IoT frameworks to establish a foundation for remote patient oversight. These early iterations primarily addressed the challenge of remote scheduling, with researchers like [3] exploring the effectiveness of audio-visual prompts (AVRD) to improve patient compliance through behavioural

cues. While successful in a laboratory setting, these systems were frequently limited by their reliance on local processing power and lacked a standardized method for transmitting adherence data across long distances or over unstable networks.

The mid-period of research shifted toward the integration of diverse sensor technologies and cellular communication to bridge the gap between the patient and the provider. [4] demonstrated the utility of combining GSM modules with ultrasonic sensors to monitor a patient's proximity to their medication. Around the same time, [5] examined Bluetooth Low Energy (BLE) as a low-power solution for localized scheduling, while [6] focused on hardware-level security, introducing biometric authentication and climate control to safeguard sensitive drug formulations. Despite the increased complexity of these devices, the literature from this era reveals a common failure point: a dependence on event-based sensing—such as detecting motion or a door opening—which failed to provide high-resolution data regarding the actual physical removal of a

dose.

Recent academic work has moved toward multi-slot architectures and cloud-based data persistence to create a more transparent record of patient activity. [7] explored IoT units that utilized mobile applications for manual intake logging, though researchers such as [10] pointed out that manual reporting remains prone to human error. To solve this, [8] and [9] implemented Infrared (IR) sensing to automate the detection of tablet removal. However, even these modern optical methods have been critiqued by [11] for their sensitivity to environmental interference and lack of physical verification. This historical progression indicates that while the industry has perfected remote notifications and basic dispensing, a significant technical gap remains for a scalable system capable of objective mass-based detection and automated inventory management.

III. System Architecture

The architecture of this system is centered on an ESP32 microcontroller that coordinates a specialized hub-and-spoke hardware configuration. To guarantee precise scheduling, the design implements a dual-layered timekeeping strategy that pairs a DS3231 Real-Time Clock for local offline stability with Network Time Protocol (NTP) synchronization via Wi-Fi. This ensures the morning, afternoon, and night dosage windows remain accurate even if the internet connection is temporarily lost. The physical sensing is handled by three 1kg load cells interfaced through HX711 amplifiers, which are positioned beneath each medication compartment. The ESP32 continuously polls these sensors to detect weight variations, allowing it to distinguish between a successful intake and an unauthorized access event. For local feedback, the system is linked to a high frequency buzzer that serves as both a reminder and a safety interlock, remaining active during dosing errors until the patient restores the medication to its correct slot.

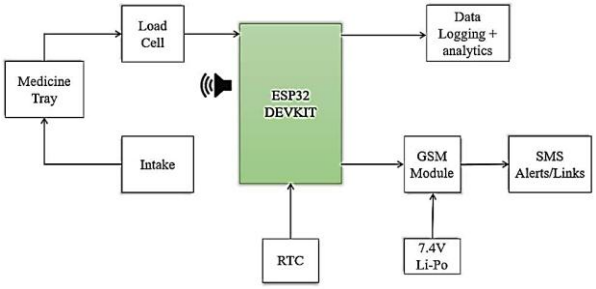


Figure 1: Proposed System Block Diagram

To ensure high reliability in data management, the system utilizes a split-path communication architecture. A SIM800L GSM module is dedicated solely to the notification layer, sending SMS reminders directly to the patient's phone to trigger the intake process. Simultaneously, the ESP32 uses its Wi-Fi capabilities to handle the cloud-logging layer. When a dose is taken or missed, the system generates a JSON payload containing the timestamp, intake status, and current inventory counts for each slot. This data is transmitted to a Google Apps Script, which populates a Google Sheets dashboard for remote viewing. This separation of tasks ensures that cellular alerts for the patient are not delayed by cloud processing, while the caregiver or physician receives a

live, globally accessible update of the "Total Left" inventory and overall adherence trends.

IV. Working Principle

The system operates through a programmed logic on the ESP32 that matches physical patient activity with a digital record. To ensure timing accuracy, the device uses a dual-check system where it pulls time from a DS3231 Real-Time Clock and periodically synchronizes with the Network Time Protocol (NTP) via Wi-Fi. When the current time matches one of the three pre-set schedules (Morning, Afternoon, or Night), the ESP32 triggers the SIM800L GSM module to send a reminder text to the patient's phone. Once this message is sent, a five-minute window begins. During this time, the system monitors three 1kg load cells via HX711 amplifiers. When the patient picks up the medication from the correct slot, the load cell detects the weight reduction, and the system registers the dose as "Taken." After the first successful dose of the day is detected, the system automatically sends a follow-up SMS to the caregiver or doctor containing a direct link to the Google Sheet where the data is being hosted.

To prevent dosing mistakes, the system includes an active safety interlock for "Wrong Compartment" detection. Even while waiting for the correct dose to be taken, the controller monitors the other two inactive slots. If a weight change is detected in a slot that does not match the current time—for example, if the patient tries to take an evening pill during the morning window—the buzzer will sound immediately. This alarm is designed to stay on until the patient puts the medication back into the correct compartment, forcing a correction of the error. If the five-minute window expires and no weight change is detected in the correct slot, the system logs the status as "Missed" and updates the record accordingly.

The logging process happens automatically over Wi-Fi, sending a JSON payload to a Google Script that updates a Google Sheet. The log includes the date, the exact time, the slot type, and whether the dose was "Taken" or "Missed." Crucially,

the system also tracks the physical inventory of the medication. Every time a dose is taken, the system subtracts one from the specific slot count (Morning, Afternoon, or Night Left) and updates the "Total Left" count. This allows the prescriber to see exactly how many pills remain in the device at any time. By tracking these counts, the system can automatically flag when the dispenser is running low, allowing for timely refills. This combination of weight-based sensing, active buzzer alerts, and inventory tracking provides a reliable way to manage medication remotely without relying on the patient to manually log their own data

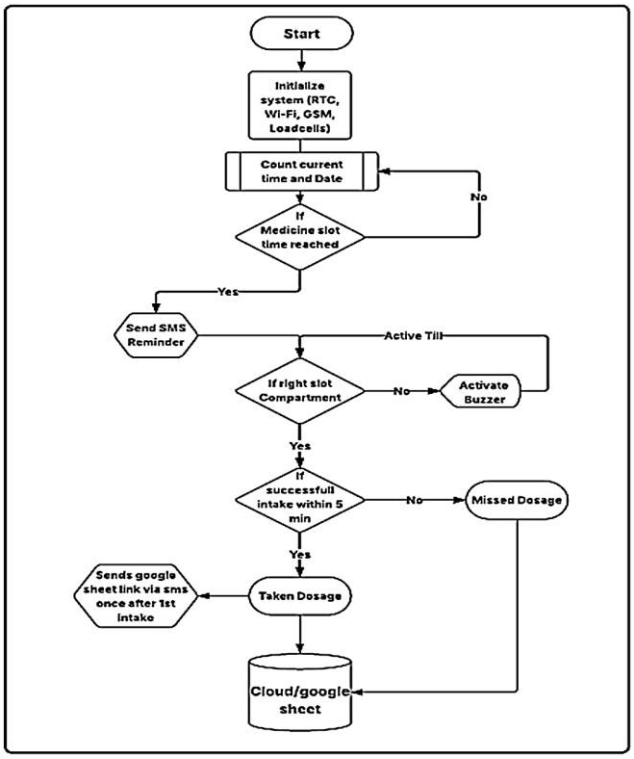


Figure 2: Working Flow of Proposed System

V. PROTOTYPE IMPLEMENTATION

The physical construction of the dispenser, as shown in the front view (Figure 3a), utilizes a vertical three-tier drawer system labelled for morning, afternoon, and night doses. This compact, stackable design allows for a clear organizational structure that is intuitive for elderly patients. The exterior houses the primary alerting components, including an LED indicator and the sound emitting buzzer, which provide local feedback during scheduled intake windows or safety violations. By using a drawer-based mechanical structure, the prototype ensures that each medication type is physically isolated, facilitating independent weight monitoring for each scheduled dose.

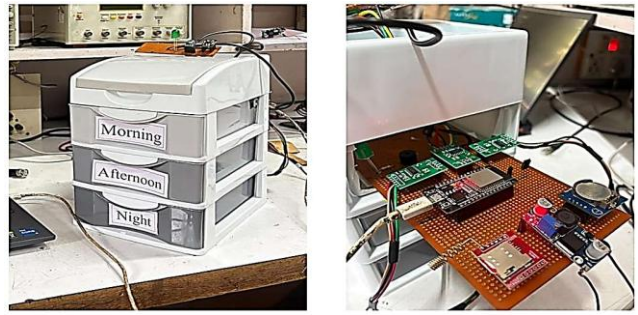


Figure 3: Dosesure Prototype

(fig 3a: Front View and fig 3b: Internal Circuitry)

A detailed view of the internal circuitry (Figure 3b) reveals the integration of the primary control and communication modules onto a customized PCB. The ESP32 DevKit occupies the central position, connected to the three HX711 amplifiers that process signals from the load cells mounted beneath the drawers. To manage the high-power requirements of the GSM module, a dedicated 7.4V Li-Po battery and voltage regulation circuit are included, alongside the DS3231 RTC for consistent timekeeping. This layout demonstrates the system's modularity, showing how the sensing hardware for weight detection is tightly coupled with the communication hardware for GSM alerts and Wi-Fi cloud logging within a single, portable footprint.

VI. RESULTS AND DISSCUSION

The effectiveness of the prototype was tested by observing its ability to detect medication removal and synchronize that data across cellular and internet networks. As shown in the mobile interface in Figure 4, the GSM-based notification system successfully initiated the intake process by delivering timespecific SMS reminders for morning, afternoon, and evening doses. A key functional milestone was reached during the first successful dose, where the system automatically sent a followup SMS containing a direct link to the Google Sheets dashboard. This feature is particularly useful for remote monitoring, as it gives caregivers and doctors instant access to the patient's records without requiring them to install any specialized software or apps

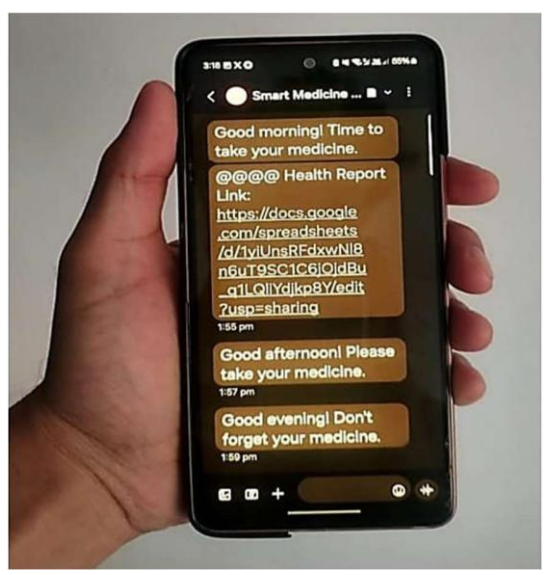
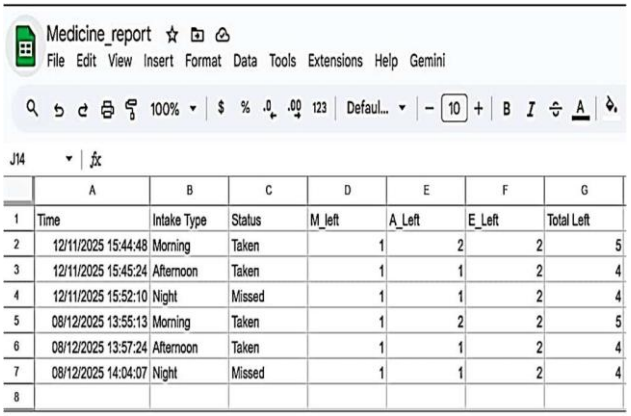


Figure 4: SMS Reminder and Google sheet link Received on Smartphone



	A	B	C	D	E	F	G
1	Time	Intake Type	Status	M_left	A_Left	E_Left	Total Left
2	12/11/2025 15:44:48	Morning	Taken	1	2	2	5
3	12/11/2025 15:45:24	Afternoon	Taken	1	1	2	4
4	12/11/2025 15:52:10	Night	Missed	1	1	2	4
5	08/12/2025 13:55:13	Morning	Taken	1	2	2	5
6	08/12/2025 13:57:24	Afternoon	Taken	1	1	2	4
7	08/12/2025 14:04:07	Night	Missed	1	1	2	4
8							

Figure 5: Google Sheet Based on Live Medicine Adherence Report

The accuracy of the weight-based sensing and cloud-logging setup is further proven by the live data captured in the digital report shown in Figure 5. The system clearly distinguished between "Taken" and "Missed" events, with the latter being recorded automatically once the five-minute experimental window closed. Beyond just tracking status, the results demonstrate that the inventory management logic works as intended. In the "Medicine report," each "Taken" entry results in an immediate deduction from the specific slot counts (M_left, A_Left, E_Left) and the "Total Left" aggregate. For example, seeing the total count drop from 5 to 4 across consecutive entries confirms that the load cells and ESP32 are precisely tracking the physical depletion of the medication. This provides a clear clinical audit trail, allowing a physician to see the exact timing of each dose and the remaining supply, which is critical for scheduling refills before the patient runs out of medicine.

VII. CONCLUSION

This project demonstrates that an IoT-based system can effectively help patients stay on track with their medication. By switching from standard sensors to high-precision load cells, the device provides a real way to verify when a tablet is actually removed. The setup uses two communication methods: a GSM module sends out text reminders, and Wi-Fi logs all the data to a Google Sheet. This keeps both the patient and the doctor updated without any data delays. Our test results show that the system accurately tracks "Taken" and "Missed" doses while also keeping an eye on how many pills are left in each drawer. Overall, this prototype offers a reliable and affordable way to monitor patients from home and ensures they get help if they fall behind on their schedule

VIII. FUTURE SCOPE

While the current version works well for tracking medication by weight, there are several ways to make it even better. A major goal for the next phase is to integrate a complete IoTbased health monitoring system using high-precision, medical grade sensors. This would allow the device to track a full range of vital signs—such as blood pressure, glucose levels, and heart rate—with clinical accuracy, rather than just acting as a pill reminder.

To make medication detection even more precise, we plan to combine IR sensors with the current load cells. This hybrid method would help identify exactly which pill was picked up, especially if a patient has more than one type of medicine in a single drawer. We also plan to update the design to hold two or more tablets per compartment for patients with complex prescriptions. Another big step will be developing a dedicated mobile app for both the doctor and the patient. This would allow a doctor to remotely set the pill names and times, while the patient or family members could easily see their health progress in real-time. Finally, we could use data analysis to spot patterns in how a patient takes their medicine, which could help predict health issues before they become serious.

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